



Bank Account Management System

A SECURE, CONSOLE-BASED BANKING SIMULATOR IN C

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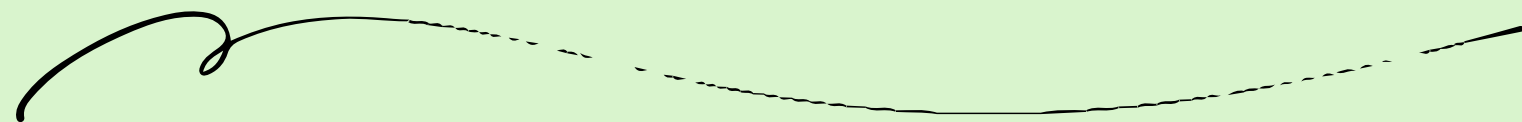
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The Problem vs. Our Solution

The Problem	Our Solution
● Beginners struggle to understand how real banking data is stored without heavy databases. ● Manual banking operations are slow and error-prone. ● Lack of offline, secure, and simple banking simulators for educational purposes. 	✓ Developed a complete offline Banking Simulator in C. ✓ Uses Binary File Handling (.dat) for permanent & secure data storage. ✓ Simulates real-world transactions (Deposit, Withdraw, Mini-Statement) instantly. 

TECHNICAL APPROACH & TECH STACK

HEADING: TECHNICAL ARCHITECTURE & TECH STACK

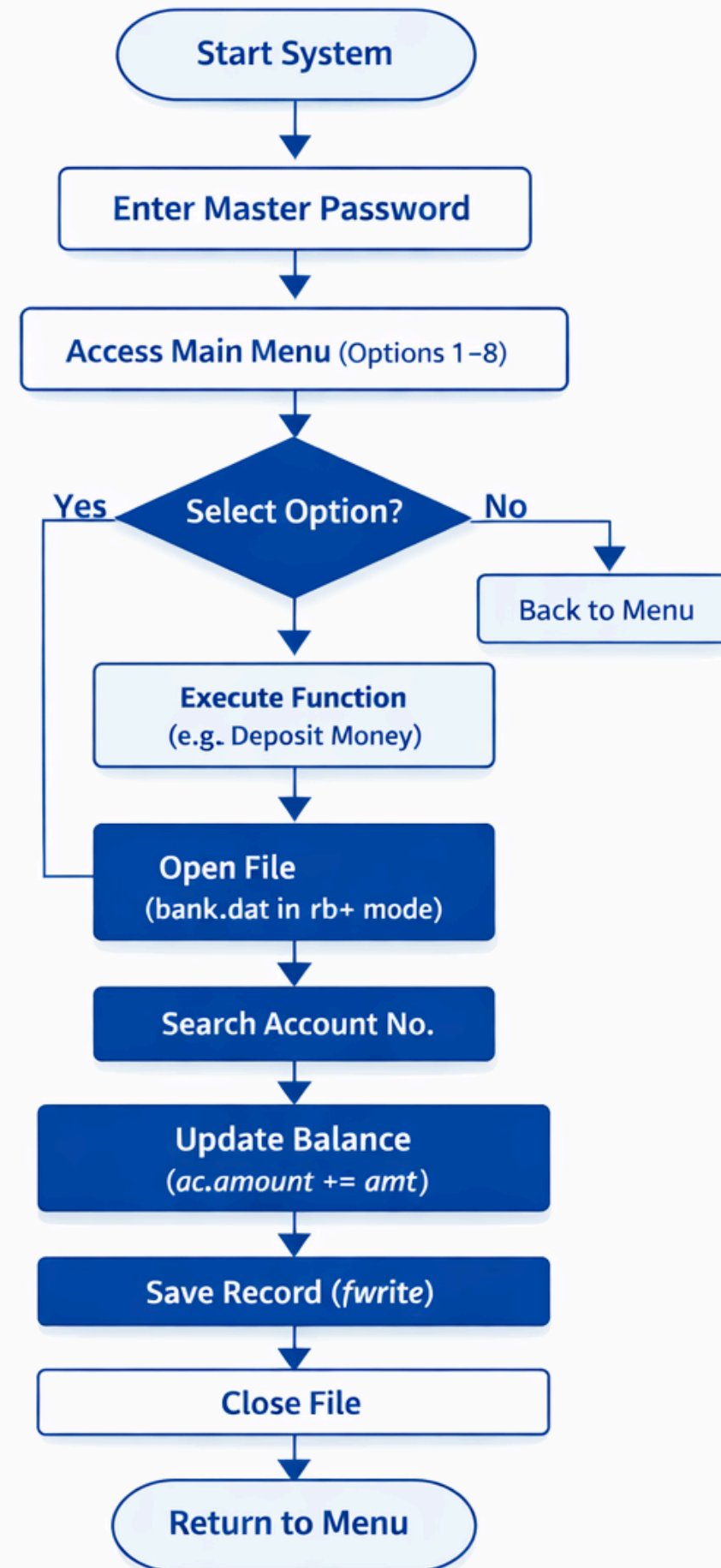
- **CORE LANGUAGE:** C PROGRAMMING (GCC / TURBO C++)
- **DATA STRUCTURES:** STRUCT (TO GROUP ACCOUNT DETAILS LIKE NAME, AC_NO, BALANCE)
- **DATA STORAGE:** NATIVE FILE HANDLING (BANK.DAT)
- **FILE OPERATIONS USED:** FWRITE() (WRITE), FREAD() (READ), REMOVE() & RENAME()
- **CORE CONCEPTS IMPLEMENTED:** POINTERS, MODULAR FUNCTIONS, LOOP CONTROL, AND TIME.H FOR TIMESTAMPS.

CORE SERVICES PROVIDED (FEATURES)

HEADING: CORE BANKING SERVICES OFFERED

1. **NEW ACCOUNT CREATION:** SECURELY OPEN AN ACCOUNT WITH INITIAL DEPOSIT.
2. **CASH DEPOSIT & WITHDRAWAL:** REAL-TIME BALANCE UPDATE WITH "INSUFFICIENT FUNDS" VALIDATION.
3. **BALANCE ENQUIRY:** INSTANTLY CHECK CURRENT ACCOUNT BALANCE.
4. **MODIFY ACCOUNT:** UPDATE NAME OR MOBILE NUMBER EASILY.
5. **CLOSE ACCOUNT:** PERMANENTLY DELETE ACCOUNT RECORDS.
6. **MINI STATEMENT:** TRACK TRANSACTION HISTORY WITH DATE & TIME.
7. **ADMIN VIEW:** DISPLAY ALL ACTIVE BANK ACCOUNT HOLDERS.

System Flow & Logic Architecture



KEY FUNCTIONS

CORE CODE FUNCTIONS & LOGIC

FUNCTION NAME	TECHNICAL LOGIC USED
CREATEACCOUNT()	USES FWRITE(&AC, sizeof(AC), 1, FP) TO APPEND NEW STRUCT DATA TO FILE.
DEPOSIT() & WITHDRAW()	USES FREAD() TO SEARCH ACCOUNT, UPDATES AC.AMOUNT, AND SAVES BACK.
DELETEACCOUNT()	COPIES ALL RECORDS TO A TEMP.DAT FILE EXCEPT THE DELETED ONE. RENAMES TEMP TO BANK.DAT.
DISPLAYALL()	USES WHILE(FREAD(...)) LOOP TO PRINT ALL RECORDS SEQUENTIALLY.
PASSWORD PROTECTION	USES GETCH() TO MASK PASSWORD WITH * AT STARTUP.

FEASIBILITY & VIABILITY

- **TECHNICAL FEASIBILITY:** 100% SUCCESSFUL. RUNS ON ANY BASIC SYSTEM WITH 0 EXTERNAL DEPENDENCIES.
- **TIME FEASIBILITY:** DEVELOPED, TESTED, AND DEBUGGED WITHIN THE GIVEN COLLEGE DEADLINE.
- **COST VIABILITY:** ₹0 COST. BUILT ENTIRELY USING OPEN-SOURCE COMPILERS AND TOOLS.
- **SECURITY:** USES BINARY FILES (.DAT) WHICH ARE NOT EASILY READABLE BY NORMAL TEXT EDITORS, PROVIDING BASIC DATA MASKING.

REAL-WORLD IMPACT & EDUCATIONAL BENEFITS

- **ZERO INTERNET REQUIRED:** COMPLETELY OFFLINE, MAKING IT HIGHLY ACCESSIBLE.
- **DATABASE-FREE STORAGE:** TEACHES HOW BANKS MAINTAINED PERMANENT RECORDS BEFORE SQL/ORACLE DATABASES WERE COMMON.
- **HANDS-ON SKILL DEVELOPMENT:** MASTERY OVER C STRUCTURES, FILE POINTERS, AND MEMORY MANAGEMENT.
- **ERROR HANDLING BUILT-IN:** PREVENTS NEGATIVE BALANCES AND VALIDATES ACCOUNT EXISTENCE AUTOMATICALLY.

Future Scope & Enhancements

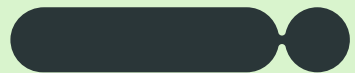
- **GUI Integration:** Moving from Console (Black screen) to a Graphical User Interface using C++ / Qt / Graphics.h.
- **ATM PIN System:** Adding individual secure PINs for every account holder.
- **Interest Calculation Module:** Automated compound interest generation for Savings Accounts.
- **Loan Management System:** Allowing users to apply for and track bank loans.
- **Database Migration:** Integrating SQLite/MySQL for handling massive user data in the future.

RESEARCH & REFERENCES

LET US C BY YASHAVANT KANETKAR (CHAPTER: FILE HANDLING).

ONLINE DOCUMENTATION: GEEKSFORGEEKS (STRUCTURES & FILE I/O).

STUDIED STANDARD OFFLINE DATA MANAGEMENT LOGIC OF LEGACY BANKING
SYSTEMS.



**THANK
YOU**



Next